Statistics.

Statistics

- **Descriptive Statistics.**
  - Generate description (Summary) of data.
- **Inferential Statistics**
  - It's used for prediction.
  - Population & Sample. eg.
  - → Sample Size, Random, Representative
  - Hypothesis Testing
  - Confidence Interval
  - Analysis of Variance (ANOVA)
  - Regression Analysis
  - Chi-Square Test.
  - Sampling Techniques.
  - Bayesian Statistics.

Types of Data.

Types of Data.

- **Categorical / Qualitative Data** (eg:) Gender.
  - **Nominal**
    - ↳ eg:) M/F
    - States → UP, MP, Bihar, UK
  - **Ordinal**
    - ↳ eg) Feedback
      - Bad
      - Avg
      - Good
- **Numerical or Quantitative Data.** eg:) Age, weight, Height
  - **Discrete Data**
    - ↳ eg) Int values only.
    - Age → 19, 18
    - Rank → 1, 2, 3
  - **Continuous Data**
    - ↳ eg) Takes any value
    - eg:) Weight → 35.8 kg.
    - Height → 5.7 ft.

Measure of Central Tendency.

Represents typical or central value of dataset.

- Mean
- Mode
- Median

Mean → Sum of all values / no. of values.

$$\text{mean} = \frac{1+2+3+4+5}{5} = 5$$

Population Mean

$$\mu = \frac{\sum_{i=1}^{N} x_i}{N}$$

N = No. of items in population

Sample Mean.

$$\bar{X} = \frac{\sum_{i=1}^{n} x_i}{n}$$

n = no. of items in the sample.

Flaw :→ Outliers can give error.

eg. $\mu = \frac{2+2+4+20}{4} = 7$

Mean shifted to outlier's side.

Median → middle value in the dataset when data is arranged in order.

eg.1) 1, 2, 3, 4, 5

Median

eg.2) $1, \boxed{2, 3}, 4$

$= \frac{2+3}{2}$

$= 2.5$ → Median.

Mode → Value that occurs frequently in dataset.

e.g.) 1, 2, 3, 1, 1, 8, 1, 2, 1

Mode = 1

e.g.) Categorical data → e.g.) Num. of Students coming from different states.

Weighted Mean.

$\mu = \frac{1+2+3}{3} = 2.$

Weighted mean = weight $\boxed{1}$ weight $\boxed{2}$ weight. $\boxed{3}$

$\boxed{LR}$ 0·2 → 10L

$\boxed{RF}$ 0·3 → 15L

$\boxed{XG}$ 0·5 → 12L

$$= \frac{0{\cdot}2\times10 + 0{\cdot}3\times15 + 0{\cdot}5\times12}{0{\cdot}2 + 0{\cdot}3 + 0{\cdot}5}$$

Trimmed Mean.

Removing some values, mean of other values.

some → small & large. } i.e. outliers.

e.g.) 20, 22, 23 { $\underbrace{25, 28, 30, 32, 35}_{\text{mean}}$ } 50, 80

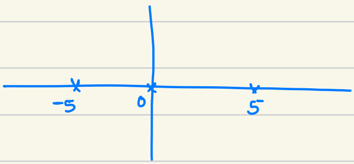
← 20% → Trimming Percentage

← 20% → Trimming Percentage.

Measure of Dispersion

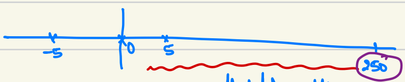
Tells the spread of the data.

Range : Min & Max value



$$\text{Range} = 5 - (-5)$$
$$= 10$$

Flaw : Affected by outliers.



affected by outlier

Variance → the avg. of sq. diff. b/w each data point & mean. It measures the avg. distance of each data point from mean. It measures the avg. distance of each point from the mean & is useful in comparing the dispersion of dataset with different mean.

$$\sigma^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n}$$

$$\sigma^2 = \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + (x_3 - \bar{x})^2}{3}$$

Population Variance

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N}$$

Sample variance

$$s^2 = \frac{\sum (x - \bar{x})^2}{n-1}$$

Mean Absolute Deviation.

$$MAD = \frac{\sum |x_i - \bar{x}|}{n}$$

↳ Not so prone to outliers.

Standard Deviation → sq. root of variance.

S.D → unit → same as of data.

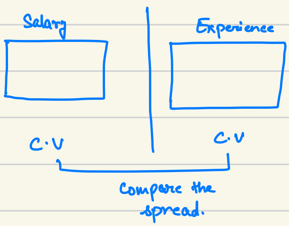
eg.) $\dfrac{(13-15)^2 + (14-15)^2 + (15-15)^2}{3} \rightarrow (lpa)^2$.

But $S.D = \sqrt{(lpa)^2}$.

$= lpa$.

Coefficient of Variation →

$C.V = (S.D / mean) \times 100\%$



Graphs for Univariate Analysis.

Categorical - Frequency Distribution Table & Cumulative Frequency.

Frequency Distribution Table. Bar Chart.

Type of Vacation	Frequency.
Beach	60
City	40
Adventure	30
Nature	35
Cruise	20
Other	15

Relative Frequency → % of category of dataset in sample. Pie Chart.

Type of Vacation	Frequency.	Relative frequency
Beach	60	0·3
City	40	0·2
Adventure	30	0·15
Nature	35	0·175
Cruise	20	0·1
Other	15	0·075

Cumulative Frequency → Running total of frequency of variable Line Chart.

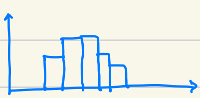
Type of Vacation	Frequency.	C.F
Beach	60	60
City	40	100
Adventure	30	130
Nature	35	165
Cruise	20	185
Other	15	200

Numerical – Frequency Distribution Tables & Histogram

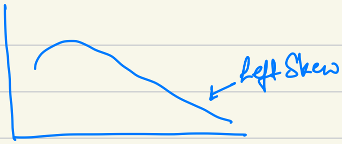
Age
30
28
41
42

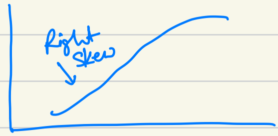
Categories → bins
↳ buckets.

0–10	11–20	21–30	31–40	41–50
		28 30		42 41



Histogram





Graphs for Bivariate Analysis.

- Categorical Categorical
- Numerical Numerical.
- Numerical Categorical

Categorical Categorical ↓

Contingency Table | Crosstab → Used to summarize the relationship b/w categorical values. Bar Chart.

Titanic.

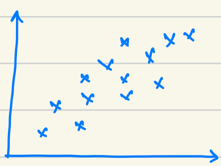
Survive	Pclass
0	1
1	2
	3

Contingency Table I

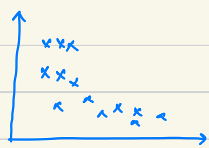
	1	2	3
0	42	31	63
1	71	118	13

↳ In Pclass 1, no. of passengers not survive → 42.

Numerical Numerical. → Scatter Plot.



+vely co-related data



-vely co-related data.

Categorical Numerical. → Bar Chart

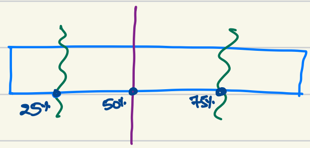
Session → 2.

Quantiles & Percentiles.

Quantiles → Statistical measures used to divide a set of numerical data into equal-sized groups, with each containing an equal no. of observations.

Use → Measure of variability, summarize & compare different datasets, also use to find outliers.

Quartiles → Q1 (25% tile), Q2 (50%) ↳(mean), Q3 (75% ile)



Deciles → Divide the data into 10 equal parts, D1 (10%ile), D2 (20% ile), . . . , D9 (90%ile)

Percentiles → Divid the data into 100 equal parts

Quantiles → Divides the data into 5 equal parts.

Things to remember :

- Data should be sorted from low to high.
- You are basically finding the location of observation.
- They are not actual values in the data.
- All other can be derived from Percentile

Percentile → % of observation in dataset that fall below a particular value.

$$P.L = \frac{P}{100}(N+1)$$

PL = Desired percentile value location.

N = Total no. of observation in the dataset.

P = the percentile rank.

eg.) Find 75%ile?

78, 82, 84, 88, 91, 93, 94, 96, 98, 99

→ Sort the data.

$$P.L = \frac{75}{100}(10+1)$$

$$P.L = \frac{33}{4} \simeq 8.25.$$

8th   9th.

→ 96 ——— 98

75% value.

$\Rightarrow 96 + (98-96) \times 0.25$

$\Rightarrow 96 + 0.25\,(2)$

$\Rightarrow 96.5$ Ans.

eg.2) Find the percentile of value 88?

$$\text{Percentile Rank} = \frac{x + 0.5Y}{N}$$

x = no of value below the given value.

Y = no. of values equal to the given value.

N = total no. of values in the dataset.

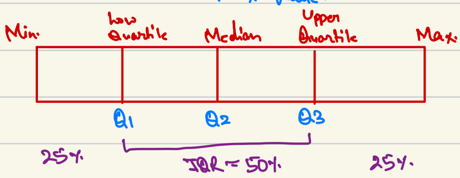
$$P.R = \frac{3 + 0.5 \times 1}{10}$$

$= 0.35 \times 100$

$\simeq 35$%ile.

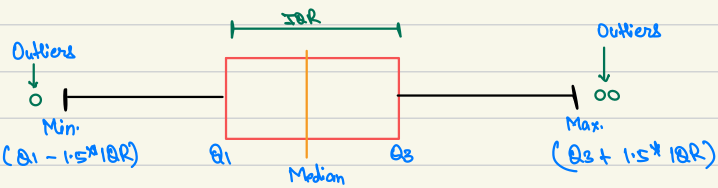
5 number summary. → Descriptive statistics, it divides data to 5 parts.

- minimum value.
- Q1 → 25%ile
- Median → 50%ile.
- Q3 → 75%ile.
- Max value.



Interquartile Range (IQR) → Diff. b/w Q3 & Q1.

Boxplot → Also known as box & whisker plot



Benefits :
1.) Easy way to see the distribution of data.
2.) Tells about skewness of data.
3.) Can identify outliers.
4.) Compare 2 categories of data.

eg.) 6, 213, 241, 260, 281, 290, 314, 321, 350, 1500
1 2 3 4 5 6 7 8 9 10

- Sort the data.
- Create box → Q1, Q2, Q3.

$$Q_2 = \frac{50}{100}(11)$$

$$= 5.5$$

$$\Rightarrow 281 + 0.5\,(290 - 281)$$

$$\Rightarrow 281 + 0.5 \times 9$$

$$\Rightarrow 285.5$$

$$Q_1 = \frac{25}{100}(11)$$

$$= 2.75$$

$$\Rightarrow 213 + 0.75\,(241 - 213)$$

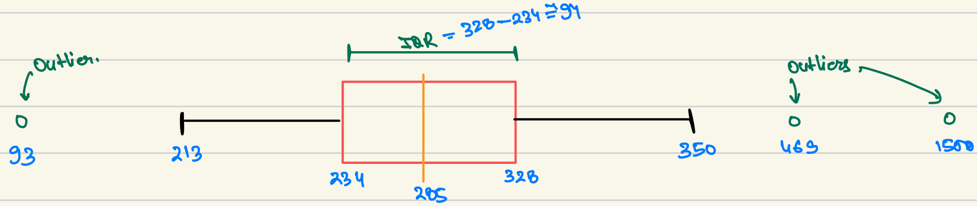
$$\Rightarrow 234$$

$$Q_3 = \frac{75}{100}(11)$$

$$= 8.25$$

$$\Rightarrow 321 + 0.25\,(350 - 321)$$

$$\Rightarrow 328.25$$



$(Q_1 - 1.5 * IQR)$

$= 234 - 1.5 * 94$

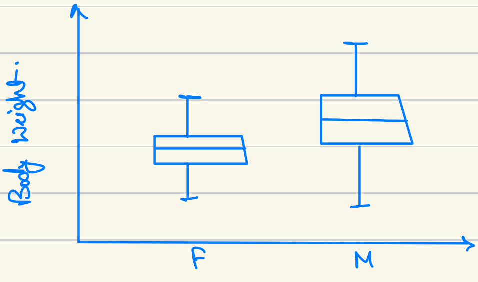
$= 93$

$(Q_3 + 1.5 * IQR)$

$= 328 + 1.5 * 94$

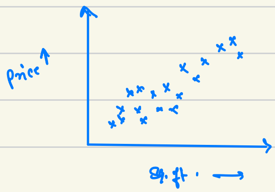
$= 469$

Side by Side boxplot



Cat weight by sex.

Scatter Plot. → 2 numerical column.



Price to Area of flats.

Covariance → Stats. measure, describes 2 variables are linearly related.
how much does 1 var. increase does var. 2 increase or decrease & to what extent.

Population

$$\sigma_{xy} = \frac{\Sigma (X - \mu_x)(Y - \mu_y)}{N}$$

X, Y = Value of x & Y in the population.
μ_x, μ_y = Population mean of x & Y.
N = Total no. of observations.

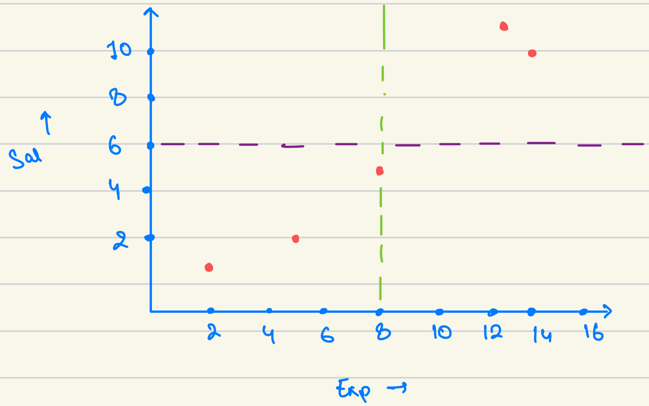
Sample

$$S_{xy} = \frac{\Sigma (X - \bar{x})(Y - \bar{y})}{n-1}$$

X, Y = Value of x & Y in the population.
$\bar{x}, \bar{y}$ = Sample mean of x & Y.
n = Total no. of observations.

9)

X	Y	X - Xmean	Y - Ymean	(X - Xmean)* (Y - Ymean)
2	1	-6	-5	30
5	2	-3	-4	12
8	5	0	-1	0
12	12	4	6	24
13	10	5	4	20
$\bar{x} = 8$	$\bar{y} = 6$			$\frac{86}{4} \Rightarrow 21.5 \longrightarrow$ +ve relationship



$$S_{xy} = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

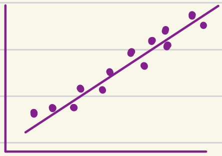
$$\Rightarrow \frac{86}{4} \Rightarrow 21.5$$

Disadvantages of Co-variance

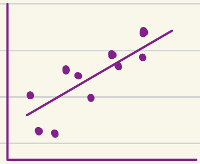
- It does not tell us about the strength of the relationship b/w two variables, since its magnitude of co-variance is affected by the scale of the variables.

Correlation

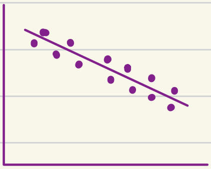
What problem does it solve?



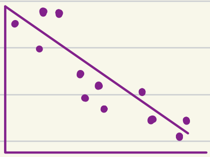
Strong +ve correlation



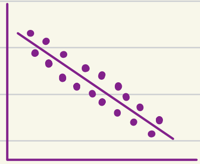
Weak +ve



Strong -ve



Weak -ve



Moderate -ve

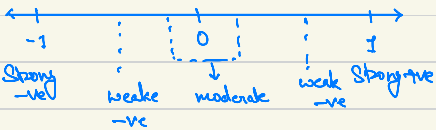
No correlation.

It measures the degree to which two variable are related and how they tend to change together.

Ranges from -1 to 1.

-1 - 0 → -ve

1 - 0 → +ve



$$\text{Correlation} = \frac{\text{Cov}(x, y)}{\sigma_x * \sigma_y}$$

St. dev.

Correlation and Causation.

Correlation does not imply Causation.

If 2 thing are co-rrelated, does not mean one thing is causing the other.
or 1 variable is reason for another variable's behaviour.